

Real Estate Data Exploration

Visual review of the model-feature dataset, with special attention to data quality, price outliers, and useful feature relationships. Every chart is kept in its own cell.

1. Load the Dataset

The path works when the notebook is started from either the repository root or the code folder.

Source: `..\data\scraped_real_estate_model_features.csv`
Rows: 1,589
Columns: 63

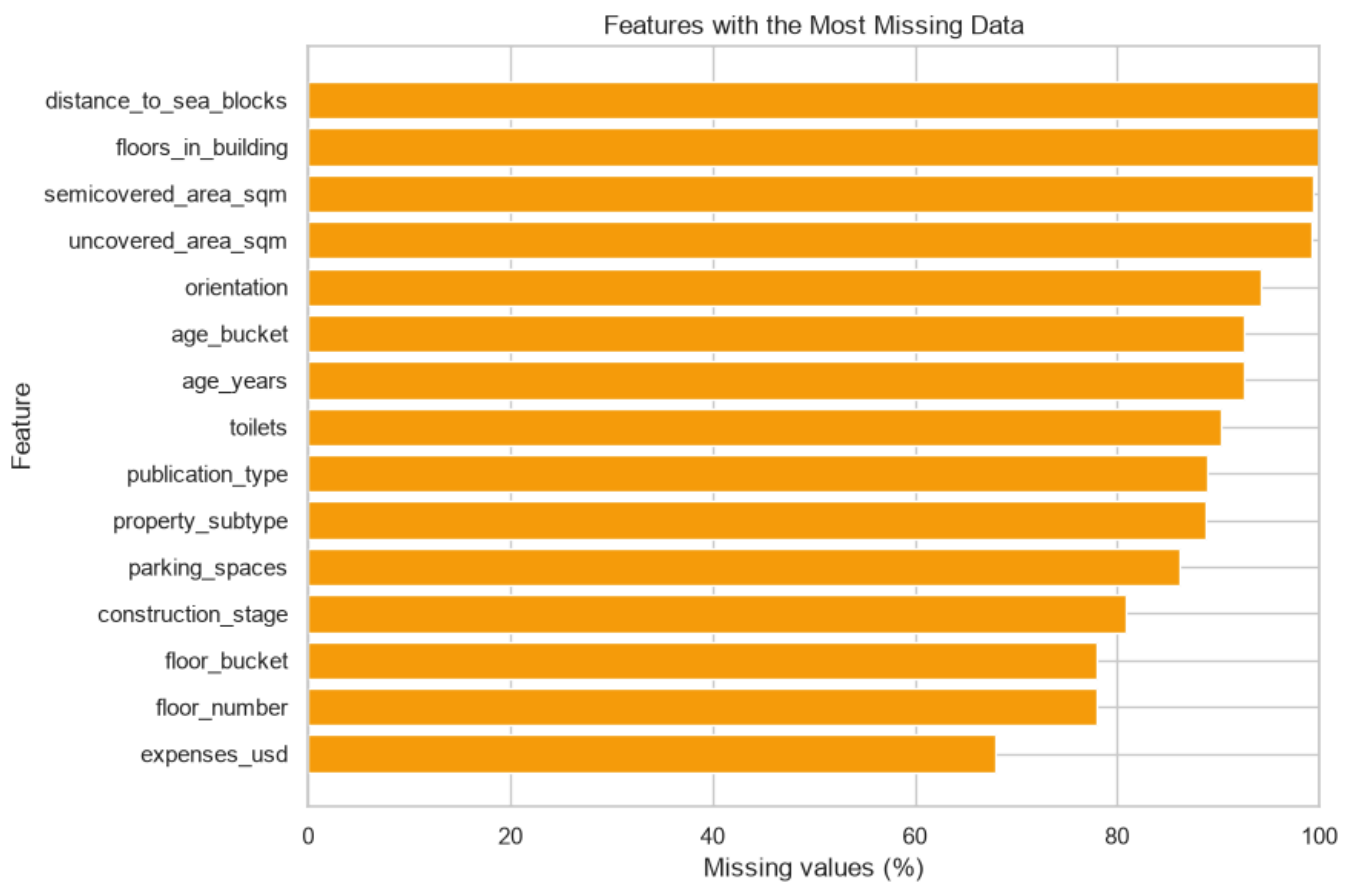
Wide raw-data preview omitted from the PDF because it is not legible on A4. The full output remains available in `code/eda_real_estate.ipynb`.

2. Data Quality

Keep the original dataframe unchanged. Use a deduplicated copy for EDA so repeated feature-and-target rows do not receive extra weight.

	value
raw_rows	1589
rows_after_feature_target_deduplication	1589
duplicate_copies	0
rows_in_duplicate_groups	0
analysis_columns	63
excluded_metadata_or_leakage_columns	0
numeric_columns	49
categorical_columns	14
all_missing_columns	2
missing_target_values	0

	missing_count	missing_percent
floors_in_building	1589	100.0
distance_to_sea_blocks	1589	100.0
semicovered_area_sqm	1580	99.4
uncovered_area_sqm	1577	99.2
orientation	1498	94.3
age_years	1471	92.6
age_bucket	1471	92.6
toilets	1434	90.2
publication_type	1413	88.9
property_subtype	1411	88.8
parking_spaces	1370	86.2
construction_stage	1286	80.9
floor_number	1239	78.0
floor_bucket	1239	78.0
expenses_usd	1081	68.0
total_rooms	1070	67.3
total_area_sqm	1059	66.6
latitude	600	37.8
longitude	600	37.8
neighborhood	585	36.8



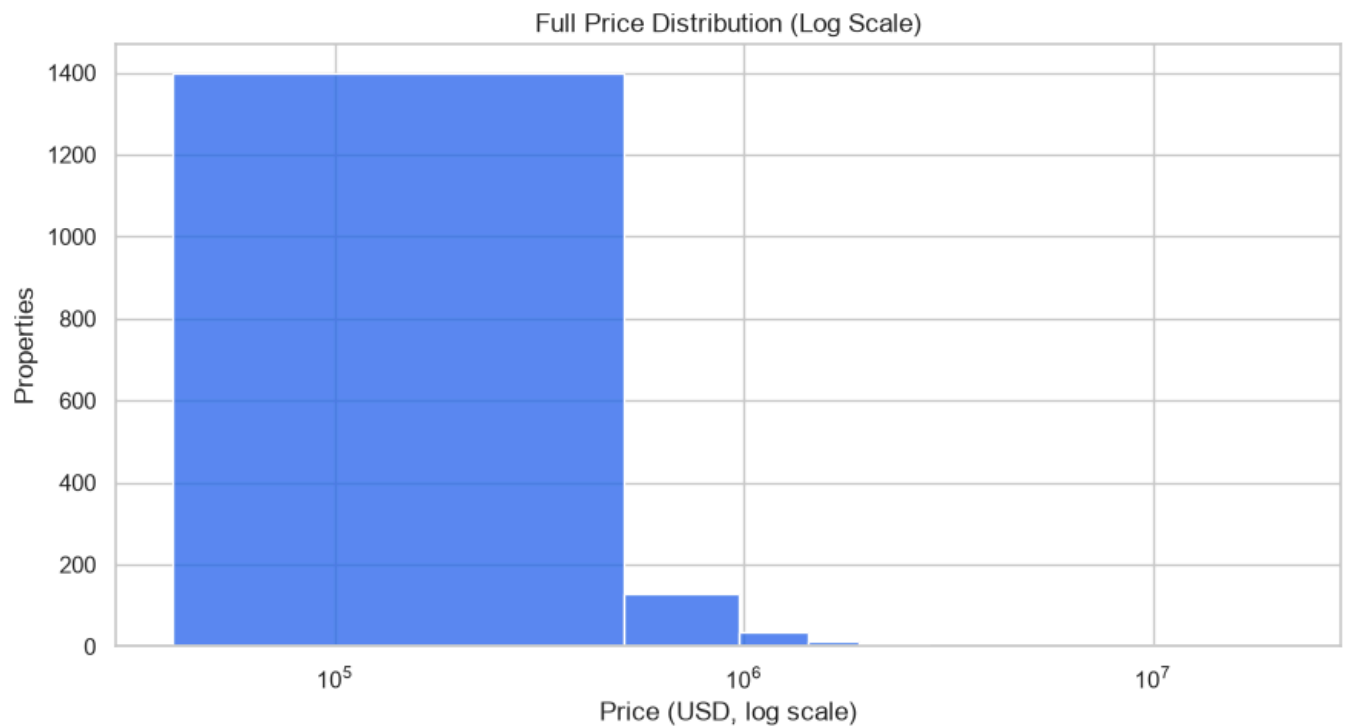
3. Price Distribution and Outliers

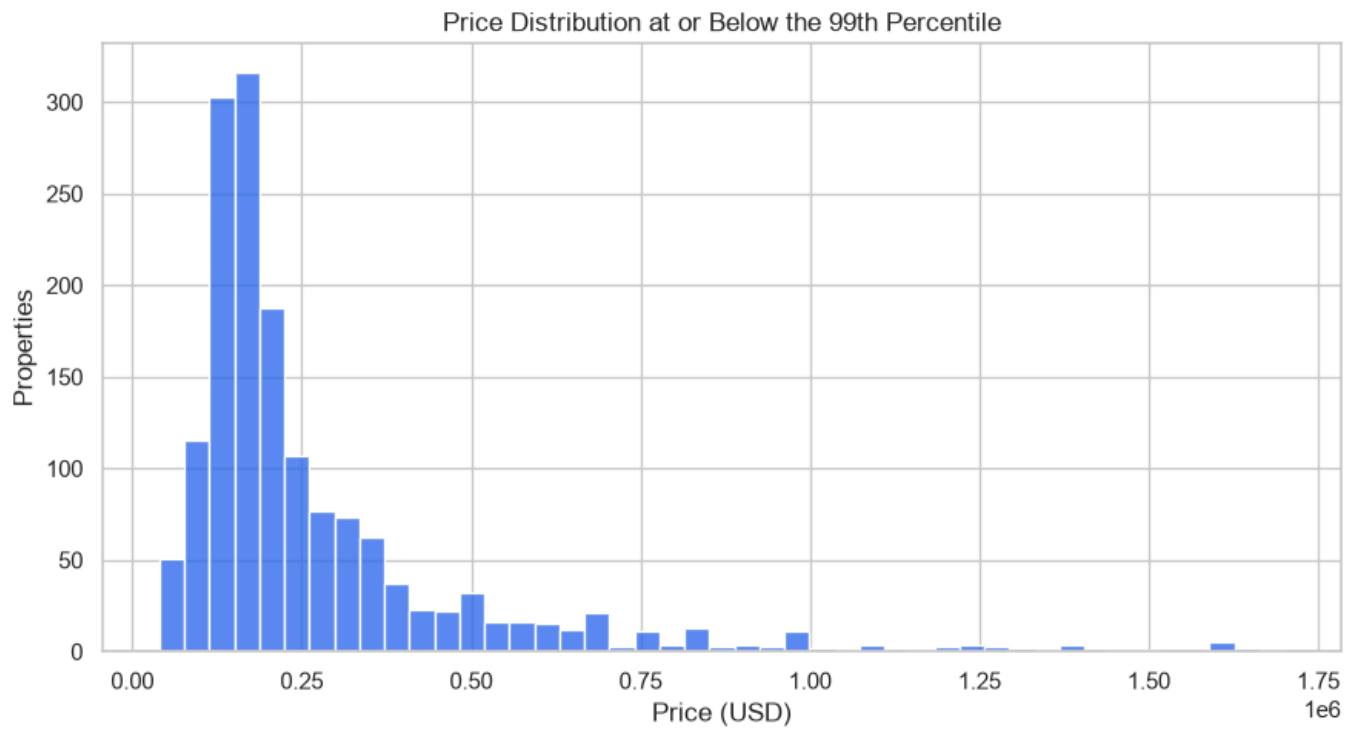
The full distribution is shown on a log x-axis. A second chart below the 99th percentile makes the main market range readable without silently deleting luxury listings.

99th-percentile review threshold: 1,706,000 USD

Rows above the threshold: 16

```
count      1589.00
mean       312186.23
std        666622.75
min         40000.00
50%        189000.00
90%        571000.00
95%        848000.00
99%       1706000.00
max       21060000.00
Name: price_usd, dtype: float64
```



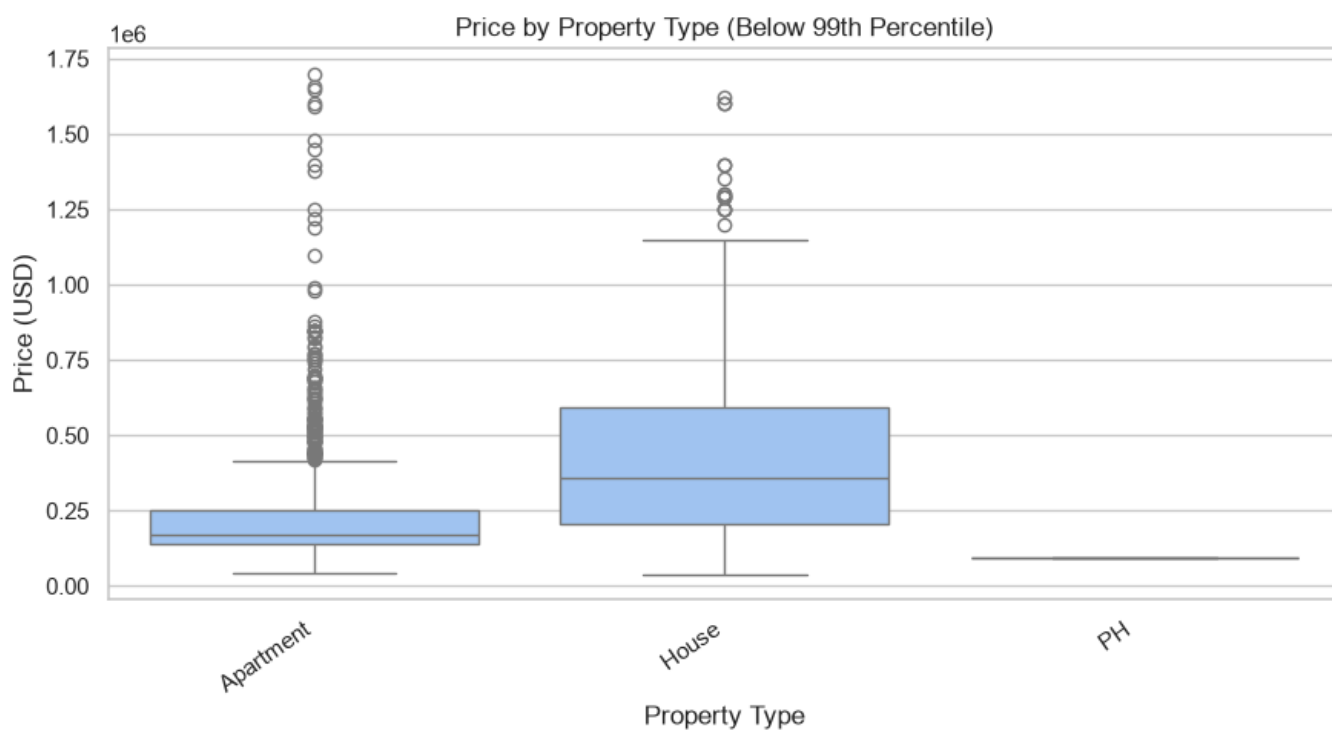
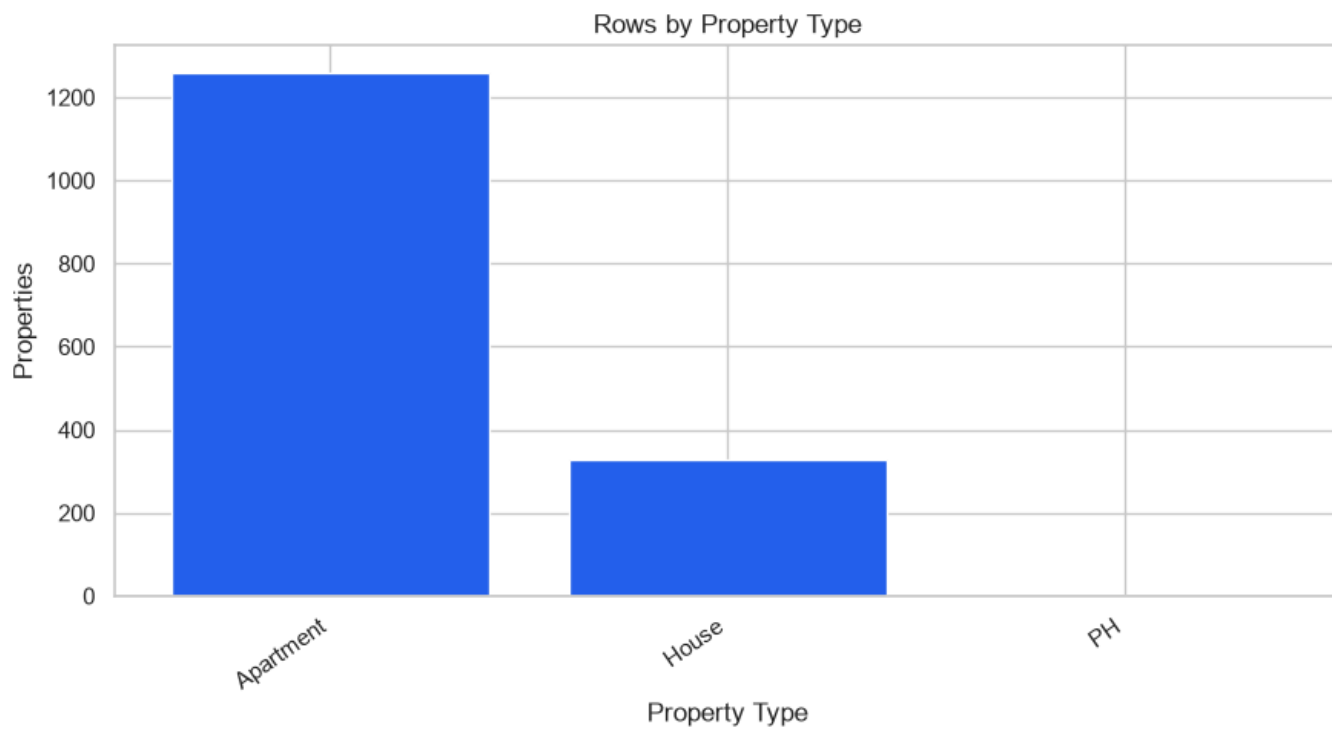


	price_usd	property_type	country	city	effective_area_sq_m	covered_area_sq_m	bedrooms	bathrooms
757	21060000	Apartment	UY	Punta del Este	NaN	NaN	6.0	8.0
756	10530000	Apartment	UY	Punta del Este	810.0	810.0	5.0	6.0
755	5460000	Apartment	UY	Punta del Este	420.0	420.0	3.0	4.0
1148	3400000	Apartment	AR	Capital Federal, Argentina,	310.0	310.0	NaN	4.0
112	2800000	Apartment	UY	NaN	250.0	250.0	2.0	2.0
1089	2600000	House	UY	NaN	800.0	800.0	1.0	3.0
1091	2600000	House	UY	NaN	680.0	680.0	4.0	7.0
1090	2600000	House	UY	NaN	800.0	800.0	1.0	3.0
30	2290000	House	AR	Troncos del Talar, Tigre, GBA Norte, Argentina	1400.0	1400.0	4.0	5.0
1094	2200000	House	UY	NaN	599.0	599.0	4.0	6.0
754	2060000	Apartment	UY	Punta del Este	203.0	203.0	2.0	3.0
1502	2000000	House	UY	NaN	567.0	567.0	5.0	4.0
1481	1800000	House	UY	NaN	687.0	687.0	4.0	3.0
1482	1800000	House	UY	NaN	510.0	510.0	4.0	4.0
1513	1750000	House	UY	Punta del Este	NaN	NaN	5.0	5.0
1561	1750000	House	UY	NaN	363.0	363.0	3.0	3.0

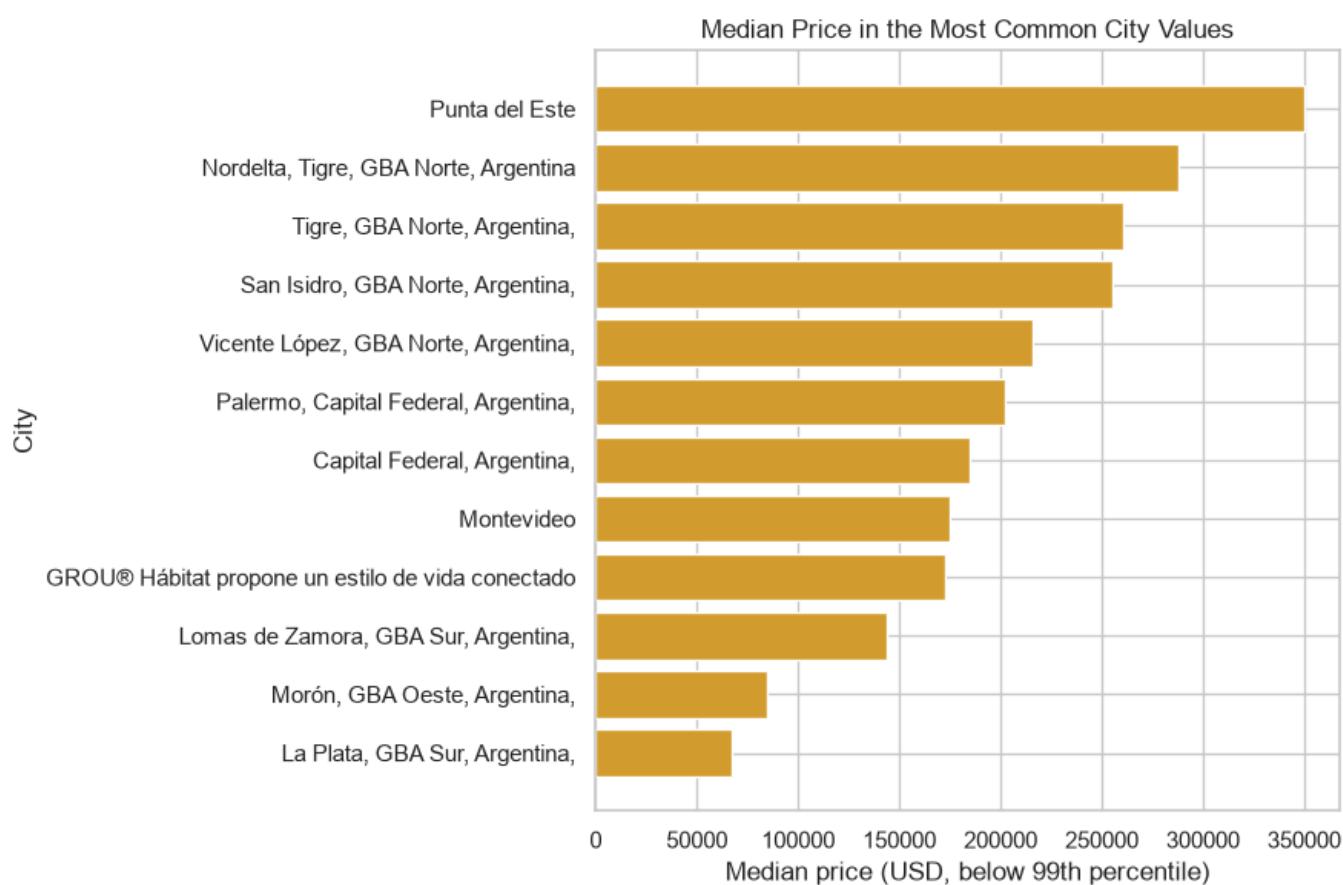
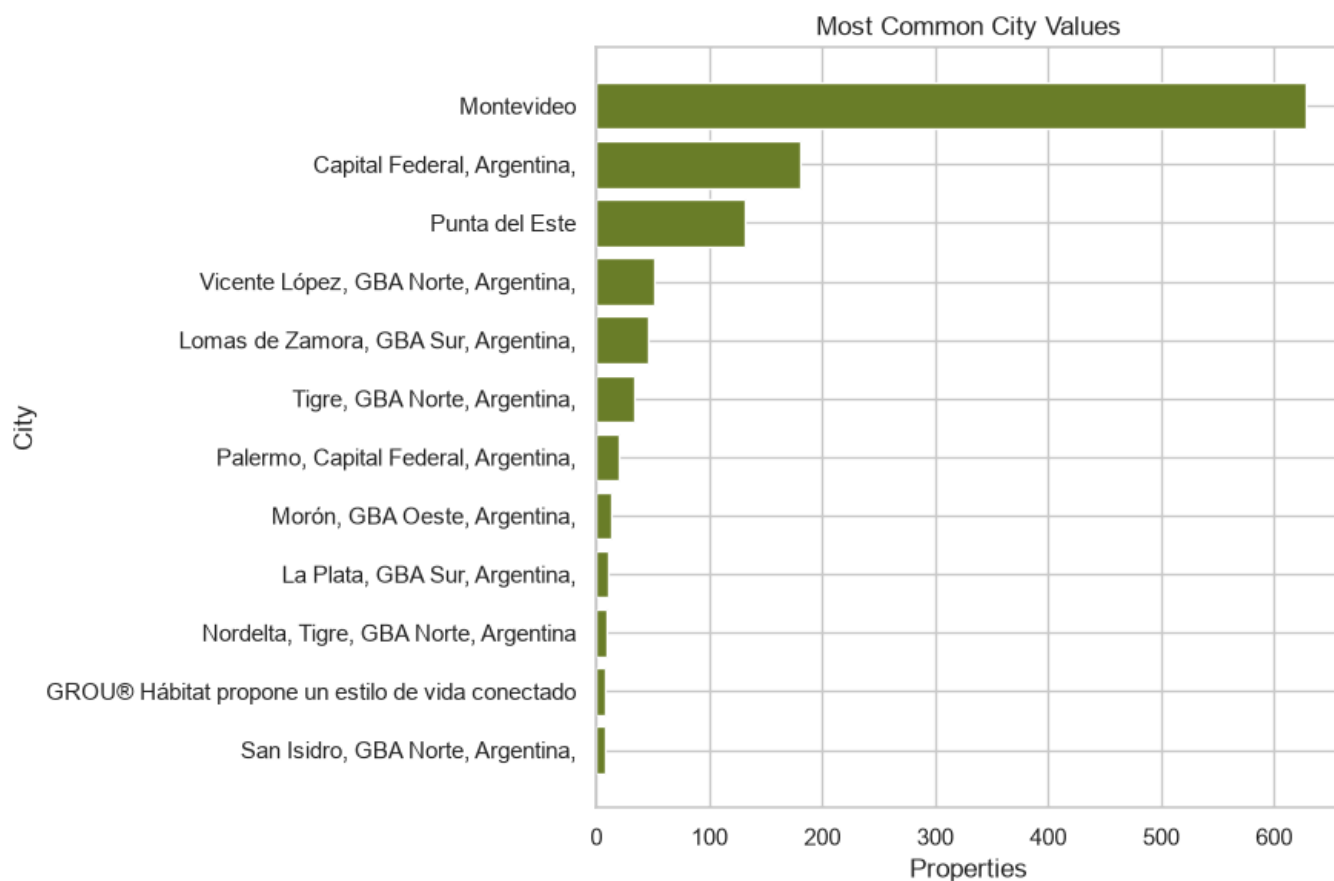
4. Property and Location Mix

Choose available categorical columns dynamically so the notebook still runs when the scraper adds or removes optional fields.

```
'property_type'
```



'city'

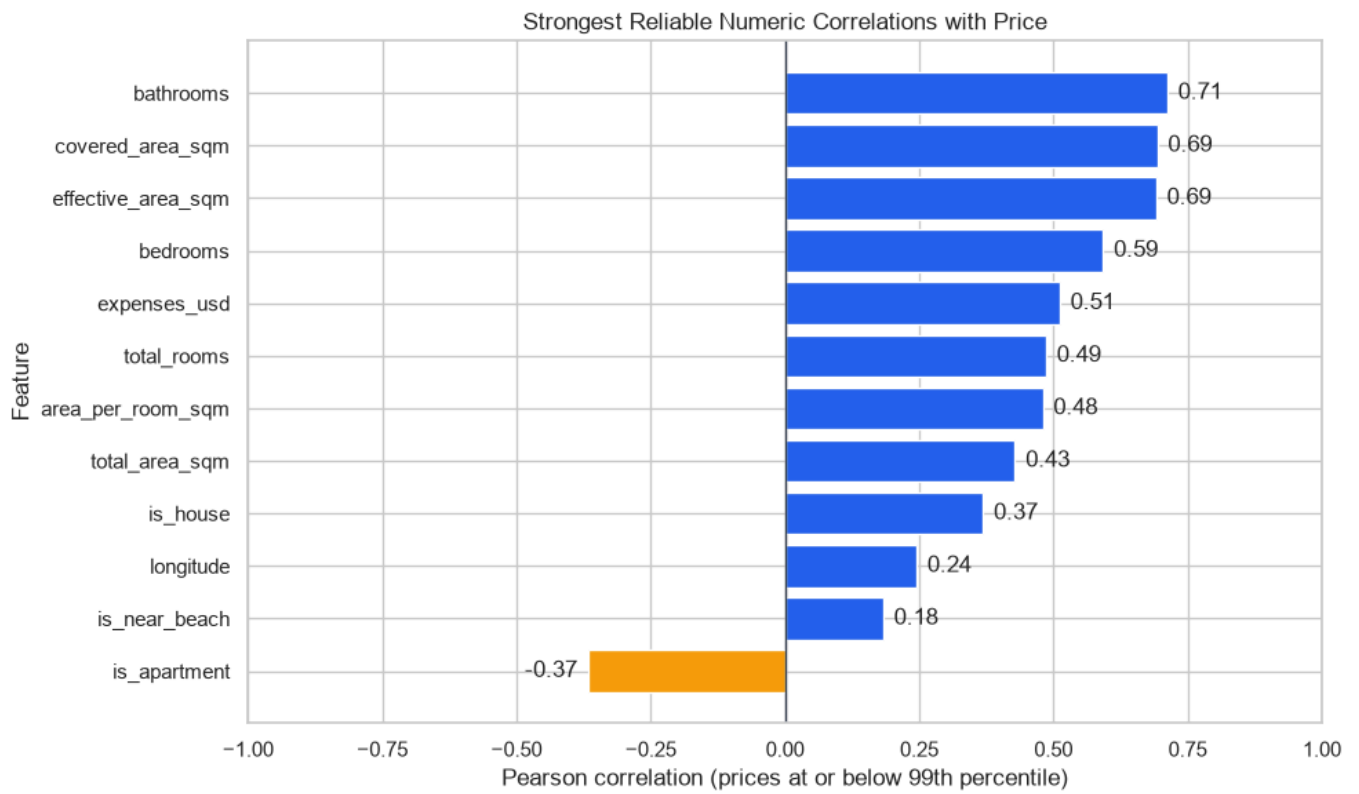


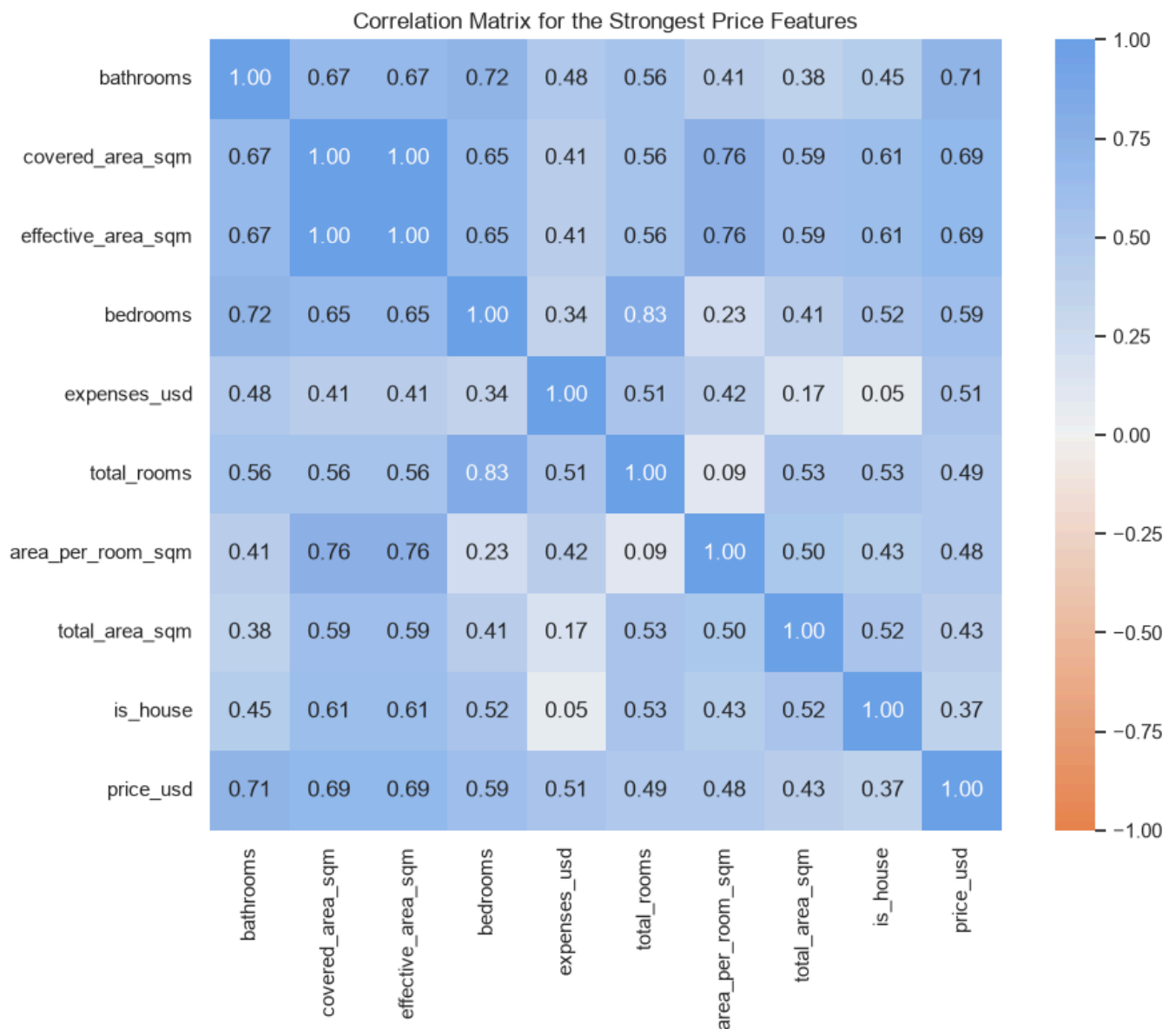
5. Price Correlations

Pearson measures linear relationships. Spearman measures ranked relationships and is less sensitive to extreme values. The table also shows Pearson correlation below the 99th price percentile and removes features with too few observed values. Correlation does not prove causation. For binary listing flags, 0 means the feature was not mentioned in the listing, not that the feature is guaranteed to be absent.

Minimum observed pairs required: 393

	pearson_all_prices	pearson_below_p99	spearman_all_prices	observed_pairs_below_p99
latitude	-0.059	-0.071	-0.207	983
longitude	0.222	0.245	0.434	983
effective_area_sqm	0.611	0.691	0.755	1564
covered_area_sqm	0.612	0.693	0.758	1561
total_area_sqm	0.460	0.428	0.624	529
area_per_room_sqm	0.468	0.480	0.468	1555
bedrooms	0.322	0.592	0.690	1558
bathrooms	0.487	0.712	0.696	1568
total_rooms	0.456	0.486	0.662	517
bathroom_ratio	0.122	0.074	-0.090	1568
has_parking	0.033	0.100	0.112	1573
is_high_floor	-0.040	-0.058	-0.034	1573
expenses_usd	0.909	0.512	0.757	501
amenity_count	-0.016	0.021	0.071	1573
is_apartment	-0.142	-0.367	-0.345	1573
is_house	0.143	0.369	0.347	1573
is_new_construction	-0.064	-0.119	-0.119	1573
is_under_construction	-0.041	-0.107	-0.156	1573
has_balcony	-0.049	-0.085	-0.080	1573
has_terrace	-0.045	-0.083	-0.000	1573

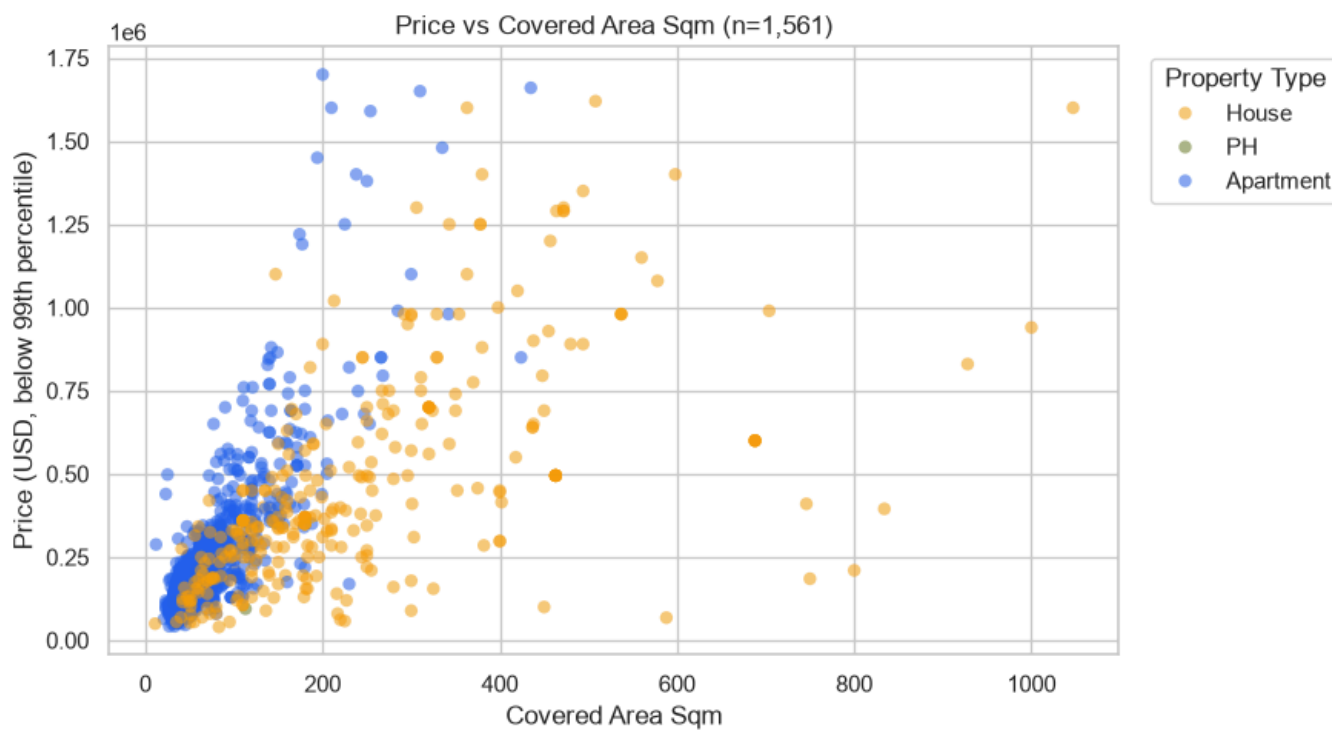
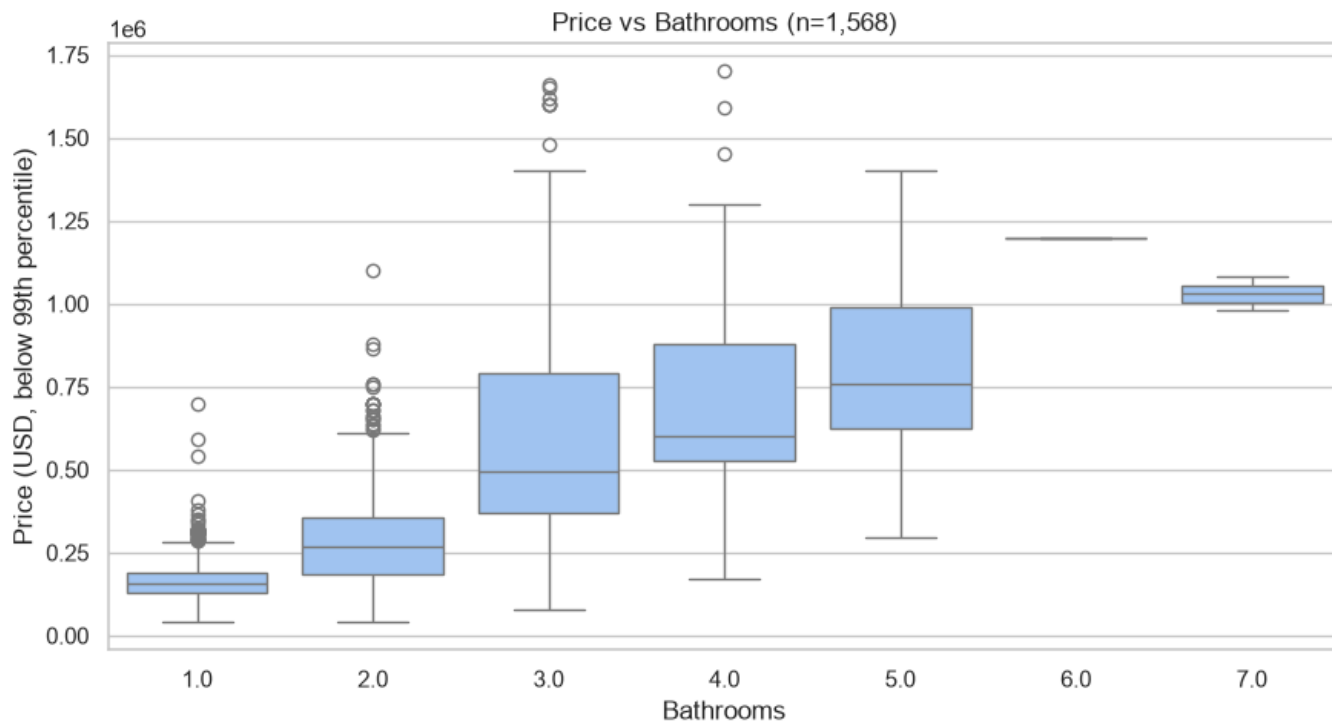


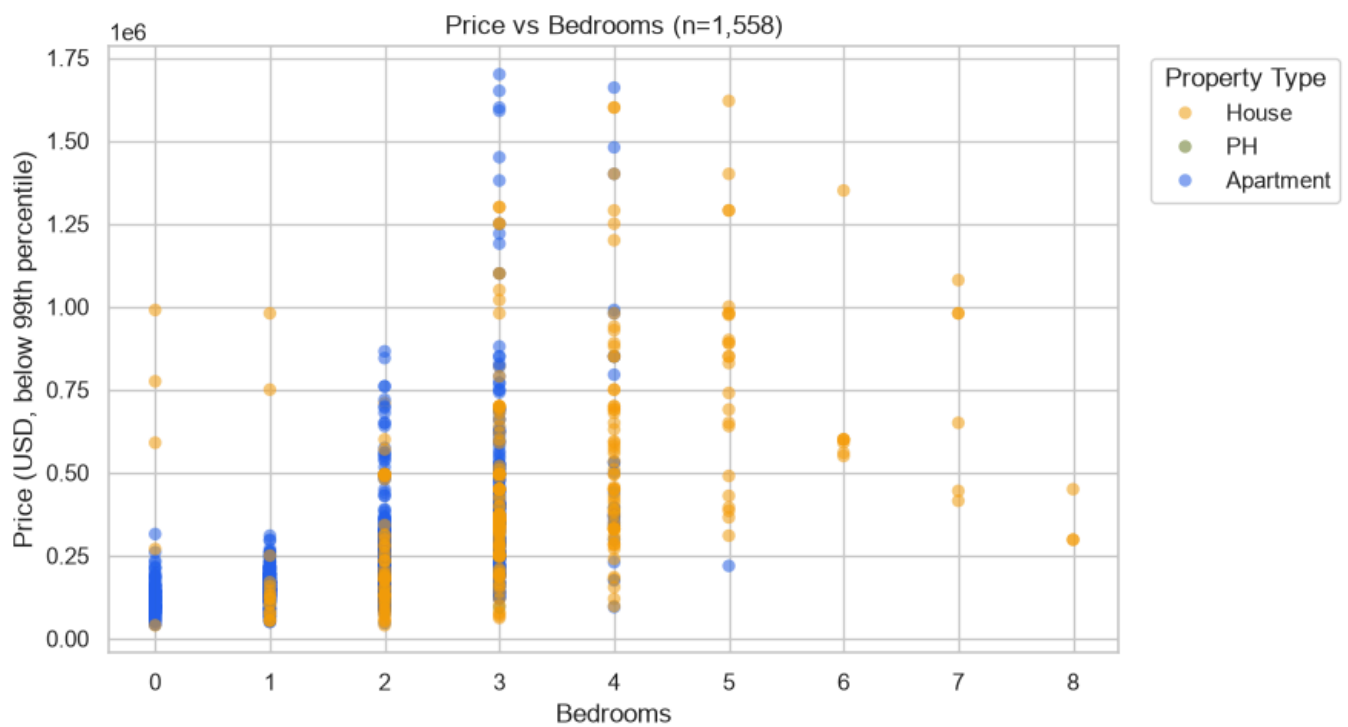
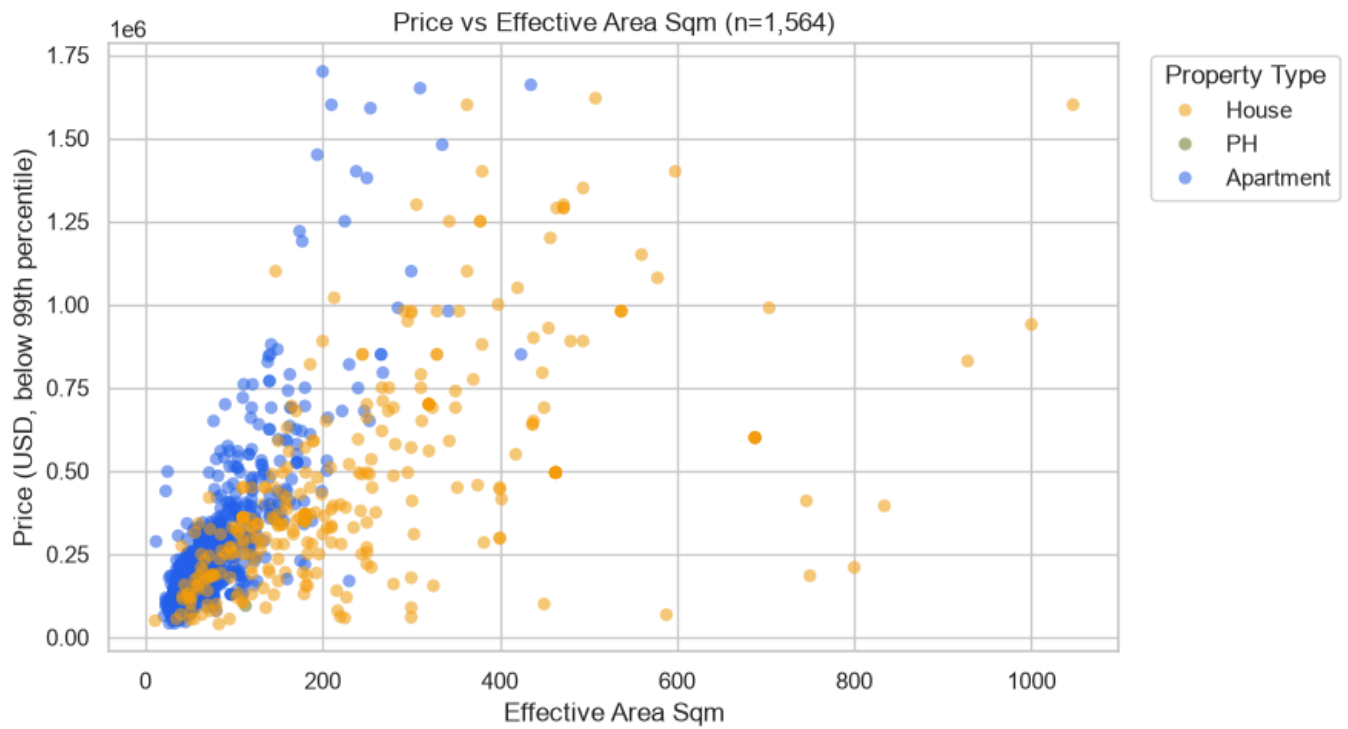


6. Strongest Feature Relationships

The four strongest reliable features are chosen automatically. Continuous features use scatter plots; low-cardinality numeric features use box plots. Each visual has its own output.

```
['bathrooms', 'covered_area_sqm', 'effective_area_sqm', 'bedrooms']
```





Takeaways

- Use the correlation table together with observed pair counts; a high correlation from sparse data is not automatically reliable.
- Review the high-price table before choosing a target transformation or an outlier policy.
- Compare Pearson and Spearman values before treating a relationship as stable.
- Binary value 0 means a feature was not mentioned in the listing, not that it is definitely absent.